

A PROJECT TITLE
ON
Spam Email Classification Using Decision Tree Algorithm and Cross-Validation Techniques

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Spam Email Classification Using Decision Tree Algorithm and Cross-Validation Techniques

1. Abstract:

Email is one of the most widely used communication methods for personal and professional purposes. However, the increasing number of spam emails has become a major concern because they waste users' time, consume storage space, and may contain phishing links or malicious content. The objective of this project is to develop a machine learning model that automatically classifies emails as spam or legitimate (ham). This project uses the Decision Tree algorithm because it is simple, interpretable, and effective for classification tasks. Before training the model, the email text is preprocessed by removing unnecessary characters, converting text to lowercase, removing stop words, and transforming the text into numerical features using the TF-IDF Vectorizer. The dataset is divided into training and testing sets, and the Decision Tree model is trained on the processed data. Cross-validation is applied to evaluate the model on different subsets of the dataset, providing a more reliable estimate of its performance and reducing the possibility of overfitting. The trained model predicts whether a newly received email is spam or not based on the learned patterns. Performance is measured using evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The developed system helps users automatically filter unwanted emails and improves email security by identifying suspicious messages. This project demonstrates the practical application of machine learning in text classification and cybersecurity. It is easy to implement, computationally efficient, and can be further enhanced using advanced machine learning or deep learning algorithms in future work.

2. INTRODUCTION: Electronic mail (email) is one of the most popular communication methods used worldwide. Every day, millions of emails are exchanged, including unwanted spam messages. Spam emails often contain advertisements, fraudulent offers that can compromise user security. The objective of this project is to develop an intelligent spam email classification system using the Decision Tree machine learning algorithm. The email content is first preprocessed and converted into numerical features using the TF-IDF Vectorizer. The Decision Tree classifier learns patterns from the training dataset and predicts whether an incoming email is spam or legitimate. Cross-validation is used to evaluate the model's performance and improve its reliability.

reduce the number of data dimensions while preserving important information, making it easier to visualize patterns, trends, and clusters.

The project includes data preprocessing, dimensionality reduction, and visualization using scatter plots. It is implemented using Python with libraries such as Pandas, NumPy, Scikit-learn, UMAP-learn, and Matplotlib. The proposed system helps users better understand environmental data and supports applications such as pollution monitoring, climate analysis, smart agriculture, and smart city development.

3. SOFTWARE REQUIREMENTS:

The following software is required to develop the project:

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.x
- IDE/Editor: Jupyter Notebook, Google Colab, or Visual Studio Code (VS Code)
- Libraries:
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn (sklearn)
 - UMAP (umap-learn)
- Dataset: Environmental Sensor Dataset (CSV format)



Signature of the Guide



Course Instructor